

UNIVERSITY OF SCHOLARS

INTEGRATED DESIGN PROJECT-1

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course Code: CSE 3108-0613

PROJECT TITLE

BAZAR-TRACE (OFFLINE-FIRST STOCK & INVENTORY SYSTEM)

1. TEAM MEMBERS :

1. Arif Hossen

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Batch: 13th

** Solo Project Presentation*

2. OBJECTIVE :

To design and implement a **zero-dependency, offline-first stock management Single Page Application (SPA)** tailored for micro-retail shops in Bangladesh.

The project aims to demonstrate advanced browser APIs (IndexedDB, Service Workers, Web Audio) integrated with an Oracle Database to keep inventory states fully synchronized and data transactions idempotent during network failures.

3. TOOLS TO BE USED :

Frontend:

- Vanilla JavaScript (ES Modules)
- Custom CSS Variables Layout
- IndexedDB

Backend & Database:

- Node.js
- Oracle Database 23ai Express
- Docker

5. TARGET AUDIENCE :

- Small-to-medium grocery shop cashiers and supervisors in connectivity-unstable regions.
- Retailers seeking lightweight stock software with zero local installation overhead.

4. KEY FEATURES :

- Offline logging:** Captures Sales & Stock receipt without active connection.
- Sync Queue:** FIFO auto uploads local operations upon reconnection.
- Native Scan:** Direct camera EAN barcode detector without third-party JS.
- Zero-library charts:** Sales line metrics rendered natively via Canvas 2D.

6. EXPECTED OUTCOME :

A fully deployable, installable Progressive Web Application (PWA) that starts instantly, enables checkout, and syncs queue elements autonomously.

The final product establishes strict role-based permission tiers (Admin / Staff) and provides clean, error-resistant SQL triggers.

PROJECT SUPERVISOR REQUESTED :

REDUANUL BARI SHOYON (SIR)